GLA UNIVERSITY, MATHURA

Department of Computer Engineering Application

SYNOPSIS

for

B.Tech CSE 3rd Year

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Intervio – AI-Powered Interview Preparation System

Overview

Intervio is an AI-driven interview simulation platform designed to help candidates prepare for real-world job interviews. The system enables users to upload their resumes or select specific technical roles, based on which it dynamically generates personalized interview questions.

A unique feature of Intervio is its **AI interviewer character**, which interacts with users by asking questions in a conversational manner. Users respond verbally, and their speech is converted into text in real time. Based on the responses, the system evaluates performance and provides intelligent feedback.

The platform also includes a **credit-based access system with online payments**, making it a scalable SaaS product. At the end of each session, users receive a **detailed performance report with visual analytics**, helping them identify strengths and areas of improvement.

🎯 Objectives

- To simulate real-world interview environments using AI
- To provide personalized interview questions based on resume or role
- To evaluate user responses and provide actionable feedback
- To help users improve communication and technical answering skills
- To build a scalable SaaS-based application with payment integration

⚙️ System Features

👤 User Functionalities

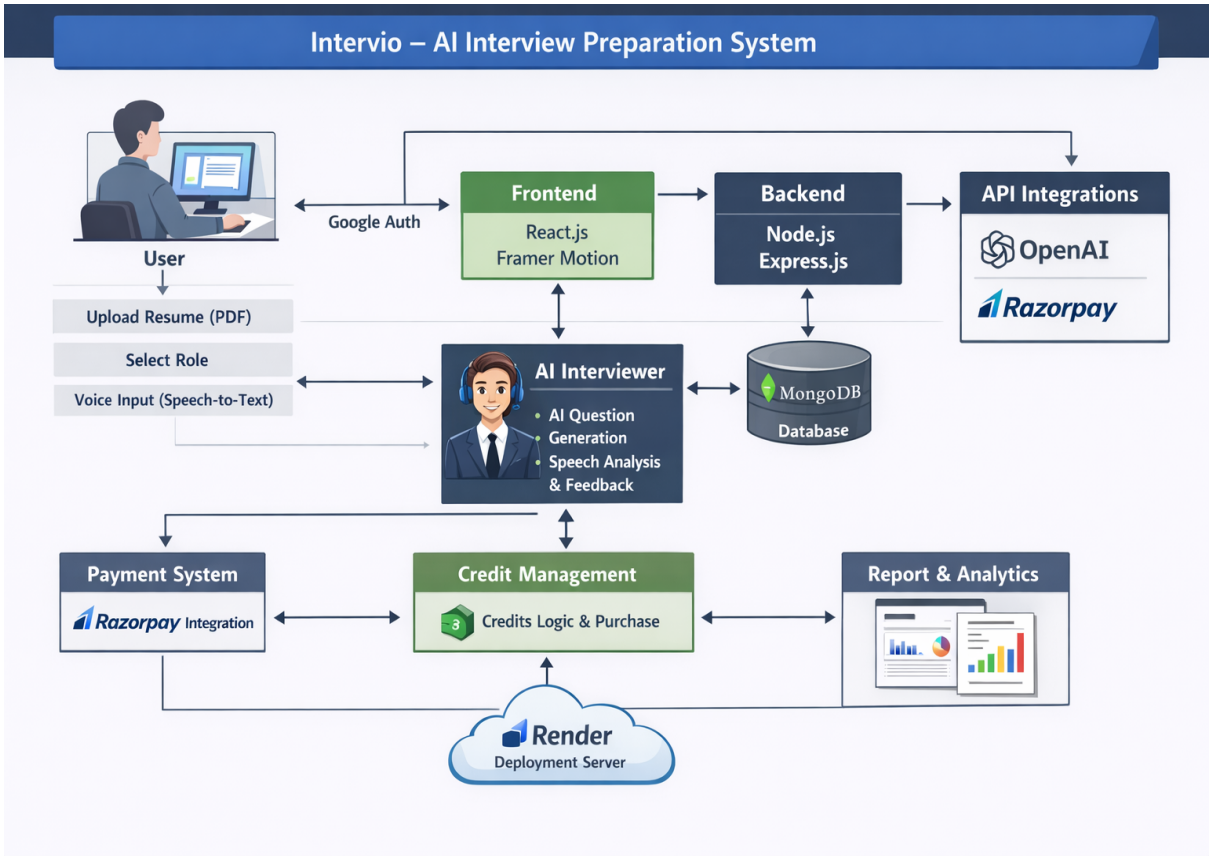
- Upload Resume (PDF) for personalized question generation
- Select interview type (Technical / HR / Role-based)
- Interact with an AI interviewer character
- Answer questions using voice input (speech-to-text conversion)
- Submit answers and receive instant feedback
- View final performance report with graphical analysis
- Access platform using a credit-based system
- Purchase credits via Razorpay payment gateway

🤖 AI-Based Functionalities

- Generate interview questions dynamically
- Analyze user responses for clarity, length, and relevance
- Provide feedback such as:
    - "Answer more clearly"
    - "Keep answers concise"
    - "Add technical depth"
- Resume-based intelligent questioning
- Time-based interview sessions (5–10 minutes)

🔄 Working Methodology

1. User logs in via Google Authentication
2. User uploads resume or selects a role
3. System generates relevant interview questions
4. AI interviewer asks questions on screen
5. User answers via voice input
6. Speech is converted into text and displayed
7. User submits the answer
8. System evaluates and provides feedback
9. After all questions, a final report is generated with graphs showing:
    - Communication quality
    - Answer relevance
    - Strengths & weaknesses



🛠️ Technology Stack

🌐 Frontend

- React.js
- Framer Motion (for smooth animations & UI)

🔧 Backend

- Node.js
- Express.js

🗄️ Database

- MongoDB

🔐 Authentication

- Firebase Google Authentication

💳 Payment Integration

- Razorpay

☁️ Deployment

- Render (Full-stack deployment)

📥 Input & Output

Input:

- Resume (PDF)
- Voice responses from user
- Selected role or interview type

Output:

- AI-generated questions
- Real-time speech-to-text answers
- Instant feedback on each response
- Final performance report with graphs

📊 Expected Results

- Improved interview confidence and communication skills
- Better understanding of strengths and weaknesses
- Realistic interview practice experience
- Scalable SaaS platform ready for real users

🚀 Future Scope

- Integration of coding editor for technical interviews
- Emotion and facial expression analysis
- Multi-language support
- AI-based resume improvement suggestions
- Company-specific interview preparation (Google, Amazon, etc.)
- Mobile application version

🎓 Conclusion

Intervio bridges the gap between theoretical preparation and real interview experience by leveraging AI and modern web technologies. It not only helps users practice effectively but also introduces them to a real-world SaaS ecosystem involving authentication, payments, and deployment.